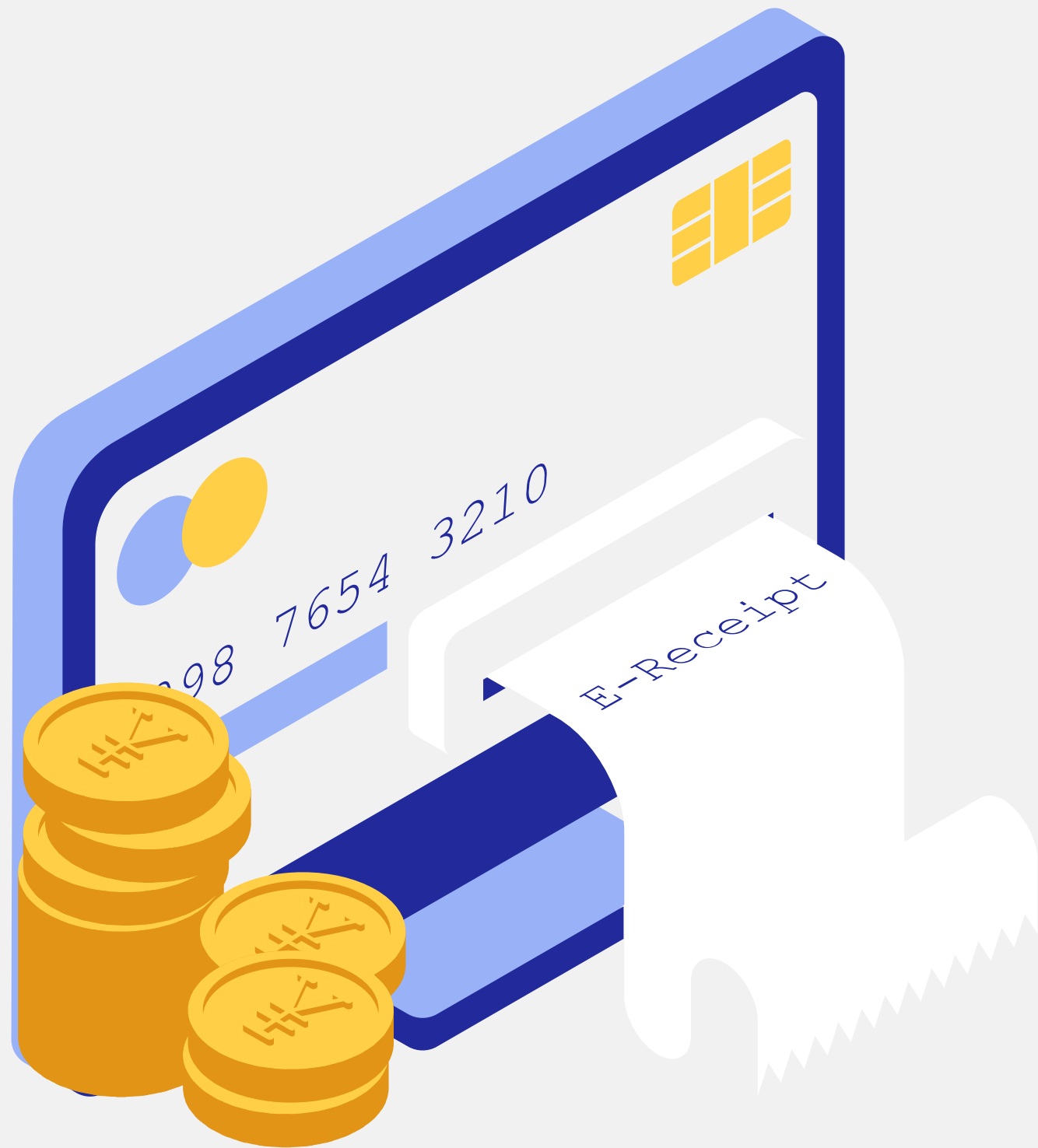


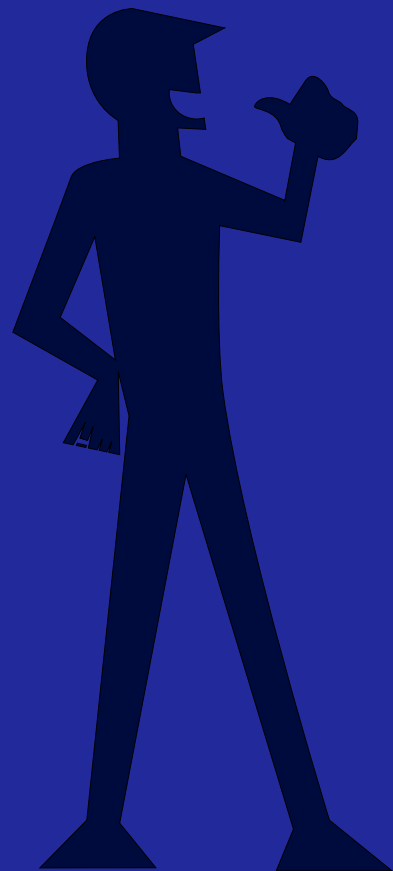


Siti Irma Dimayanti

TheLook E-commerce Business Performance Analysis

Customer, Product, Marketing & Operational Insights

Self Overview



Siti Irma Dimayanti

Aspiring Data Analyst with a background in laboratory management and data analytics. Passionate about transforming data into actionable business insights through SQL, Python, and Power BI. Currently seeking opportunities to grow as a Data Analyst in a data-driven organization.

Education

- Data Analytics & Business Intelligence Bootcamp — Dibimbing (2026)
- B.Sc. in Food Technology — Universitas Terbuka (2023)
- Diploma in Chemical Analysis — Universitas Padjadjaran (2018)

Working Experience

Laboratory Education Staff Universitas Islam Bandung (2021–Present)

- Successfully managed laboratory operations and practicum administration to support academic activities.
- Maintained accurate laboratory inventory and documentation to ensure operational efficiency.
- Improved data organization and reporting through systematic record management.

Overview

Project



✓	Project 1	Superstore Sales Dashboard (Excel)
		Developed an interactive Excel dashboard to analyze sales, profit, customer segments, and regional performance, providing insights to support business decision-making.
✓	Project 2	Sales Performance Analysis Using SQL
		Utilized advanced SQL techniques, including JOINS, CTEs, subqueries, and window functions, to analyze sales performance, shipping efficiency, product trends, and employee contributions.
✓	Project 3	Marketing Analytics Dashboard Using Tableau
		Built an interactive Tableau dashboard to evaluate marketing channel performance using CTR, CVR, revenue, conversions, and ROAS metrics for strategic decision-making.
✓	Project 4	Social Media Engagement Analysis
		Analyzed social media engagement using Python by calculating key metrics such as ER, IPR, IPI, and CTR to identify high-performing content strategies.

Overview

Project



✓	Project 5	Spotify Campaign A/B Testing Analysis
		Designed and analyzed an A/B testing experiment to evaluate campaign effectiveness using statistical hypothesis testing and business performance metrics.
✓	Project 6	RFM Customer Segmentation Analysis
		Performed customer segmentation using the RFM (Recency, Frequency, Monetary) framework in Power BI to identify high-value customer segments and support targeted marketing strategies.
✓	Project 7	Customer Churn Analysis
		Conducted exploratory data analysis using Python to identify customer churn patterns and uncover factors influencing customer retention.
✓	Project 8	Company Sales Analysis Using SQL
		Applied SQL to analyze sales performance, identify top and bottom-performing products, and generate actionable business insights from transactional data.
✓	Final Project	Retail Sales Performance & Customer Behavior Analysis
		Conducted an end-to-end retail business analysis using Python and Power BI to evaluate sales performance, customer behavior, product performance, operational efficiency, and customer satisfaction. Developed interactive dashboards and delivered data-driven business recommendations to support strategic decision-making.

Project Background

The rapid growth of e-commerce has resulted in a significant increase in transactional data across customer activities, product sales, website interactions, inventory management, and operational processes. Analyzing this data enables businesses to better understand overall performance, identify key business drivers, and support data-driven strategic decision-making.

Project Goal

- Analyze business performance across customer, product, marketing, and operational dimensions.
- Identify key drivers of business growth and operational challenges.
- Deliver actionable insights and strategic recommendations to support data-driven decision-making.

Expected Outcome

- Identify business growth opportunities.
- Improve operational efficiency.
- Strengthen customer acquisition and retention strategies.
- Enable data-driven strategic decision-making.

Key Stakeholders

- CEO
- Marketing Manager
- Merchandising/Product Manager
- Operations Manager

Business *Problem*

As TheLook continues to grow, it generates a large volume of transactional data across customers, products, marketing channels, and operational processes. However, without comprehensive analysis, identifying the key drivers of business performance, customer value, marketing effectiveness, and operational efficiency remains a significant challenge.

Business *Questions*

1. What are the key factors influencing TheLook's business performance throughout the observation period?
2. Which product characteristics contribute the most to business growth, and what differentiates high-performing products from low-performing ones?
3. How do purchasing behaviors differ across transaction frequency groups, and how do they influence TheLook's business performance?
4. At which operational stage does TheLook experience the greatest business opportunity loss, and what factors contribute to it?
5. How effective is each traffic source in acquiring high-value customers and contributing to overall business performance?

Why It *Matters*

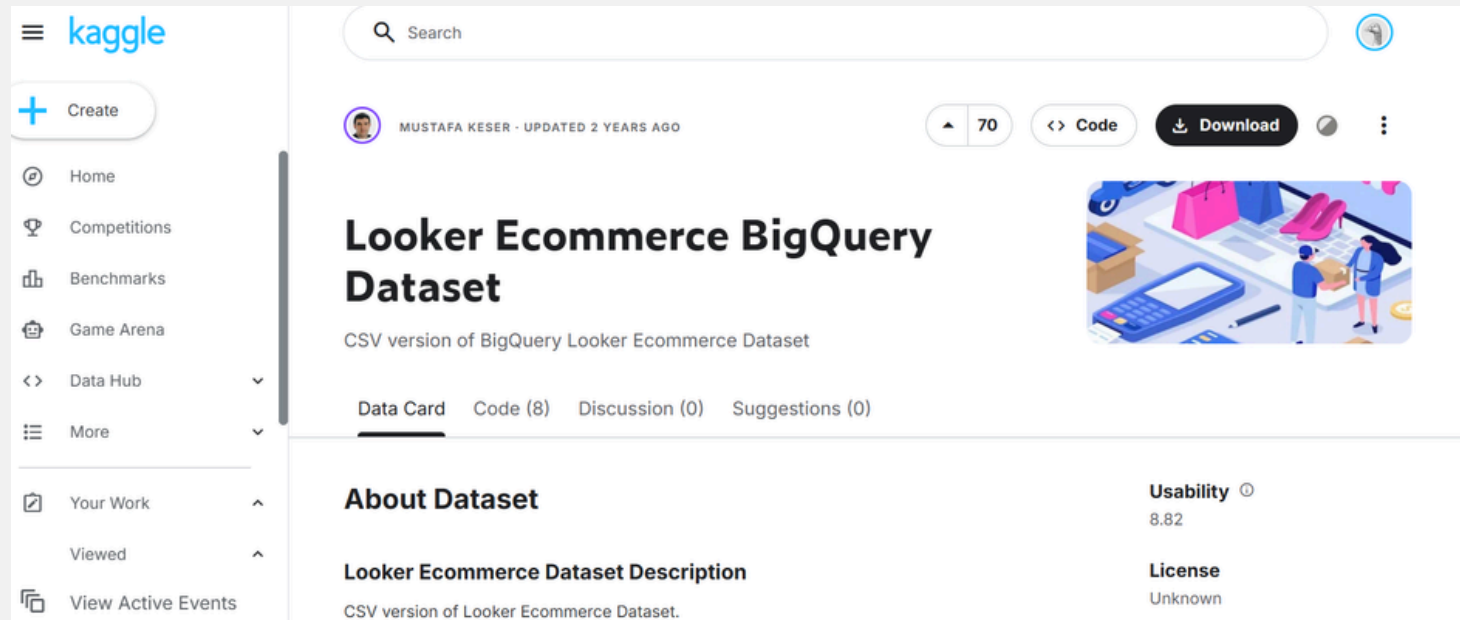
1. Understand the key drivers behind overall business performance and growth.
2. Identify high-performing products to optimize product strategy.
3. Recognize high-value customer segments to improve customer retention.
4. Detect operational bottlenecks that lead to revenue loss.
5. Evaluate marketing channel effectiveness to optimize customer acquisition and marketing investment.

Business Impact

The insights generated from this analysis enable TheLook to:

1. Develop more effective business growth strategies.
2. Optimize product portfolio management and merchandising decisions.
3. Strengthen customer retention and lifetime value.
4. Improve operational efficiency and reduce business losses.
5. Allocate marketing resources more effectively to maximize business performance.

Data Understanding

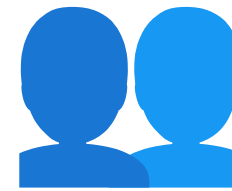


- Dataset: TheLook E-commerce Dataset (Synthetic)
- Source: Kaggle
- Collection Method: Publicly available secondary dataset
- Structure: 7 Relational Tables
- Data Type: Relational Transactional Dataset

~180K+
Records

2019-2024
Period

Key Entities



Customers | Products | Orders | Distribution | Traffic Sources

Key Variables

- Customer Demographics
- Product Information
- Transaction Details
- Order Status
- Traffic Source
- Revenue & Profit

Data Preparation

Data Quality Assessment

Assessment	Result
Missing Values	Informative missing values identified and retained
Duplicate Records	No duplicate records found
Referential Integrity	No integrity issues detected
Data Types	Successfully validated and standardized
Outliers	Valid business variations retained

Data Quality Assessment

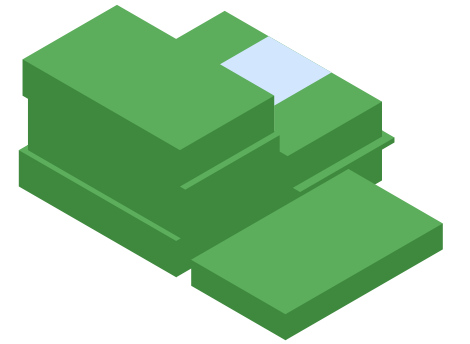
- Converted timestamp columns into datetime format.
- Created analytical features (Profit, Order Year, Order Month, Age Group).
- Integrated seven relational tables into a single analytical dataset.

Output

A clean, validated, and integrated dataset ready for Exploratory Data Analysis (EDA) and Power BI dashboard development.



Data Analytics



Analytical Techniques	This project applies multiple analytical techniques to understand business performance and answer the defined business questions.
Descriptive Analysis	Summarize overall business performance using key metrics such as revenue, profit, orders, and customers.
Distribution Analysis	Analyze the distribution of customers, products, orders, and website activities to understand data characteristics and business composition.
Trend Analysis	Evaluate business performance over time by analyzing revenue, order volume, active customers, growth rate, and average order value trends.
Comparative Analysis	Compare performance across product categories, customer segments, traffic sources, and operational metrics to identify key business drivers.
Relationship Analysis (EDA)	Explore relationships between customer behavior, product performance, marketing channels, and operational metrics to generate actionable business insights.

Insights

These insights address the identified business problems by revealing key growth drivers, customer behavior, operational challenges, and marketing effectiveness, enabling data-driven strategic decision-making.



Business Performance

Business growth was primarily driven by higher transaction volume rather than an increase in average order value, highlighting transaction growth as the main driver of business performance.



Product Performance

Outerwear & Coats emerged as the strongest revenue and profit contributor, making it a key product category for business growth.



Customer Analytics

Most customers made only a single purchase, indicating that improving customer retention represents the largest opportunity for sustainable business growth.

Insights

These insights address the identified business problems by revealing key growth drivers, customer behavior, operational challenges, and marketing effectiveness, enabling data-driven strategic decision-making.



Operational Performance

Cancelled orders accounted for the largest potential business opportunity loss, emphasizing the need to improve order fulfillment and reduce cancellations.



Marketing Effectiveness

Search was the most effective traffic source by generating the highest customer acquisition and revenue contribution, making it the strongest marketing channel.

TheLook E-commerce Business Performance Analysis

Business Growth & Product Performance Insights

Year

All

Category

All

10.8M

Total Revenue

5.6M

Total Profit

125.2K

Total Orders

80K

Active Customers

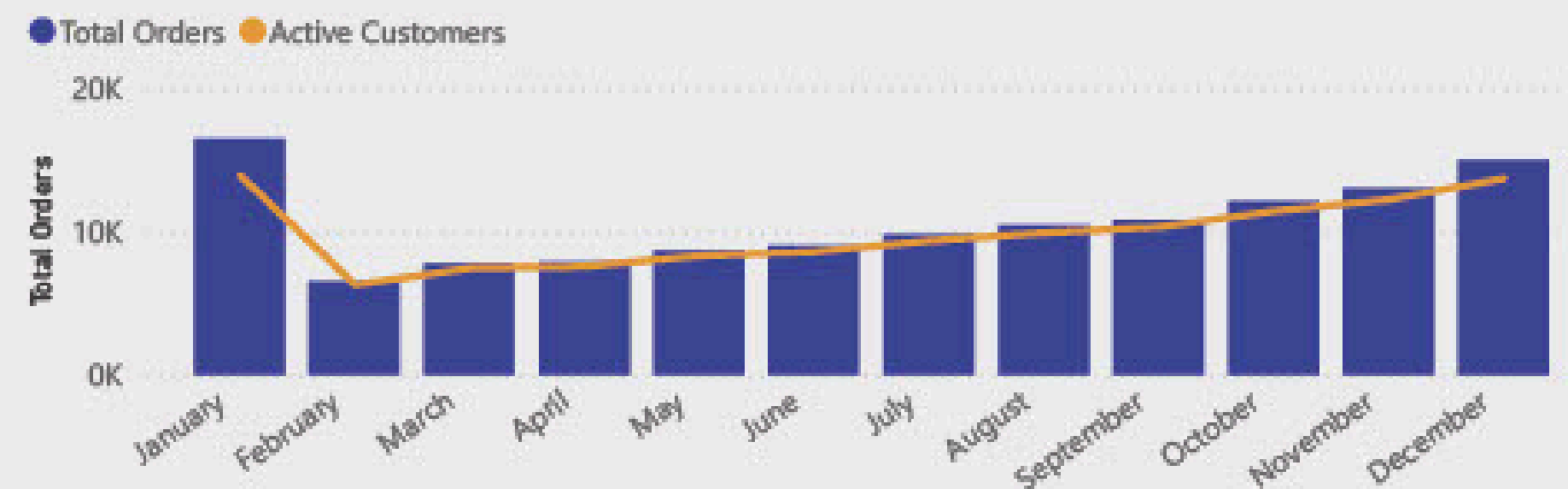
\$86.46

Average Order Value

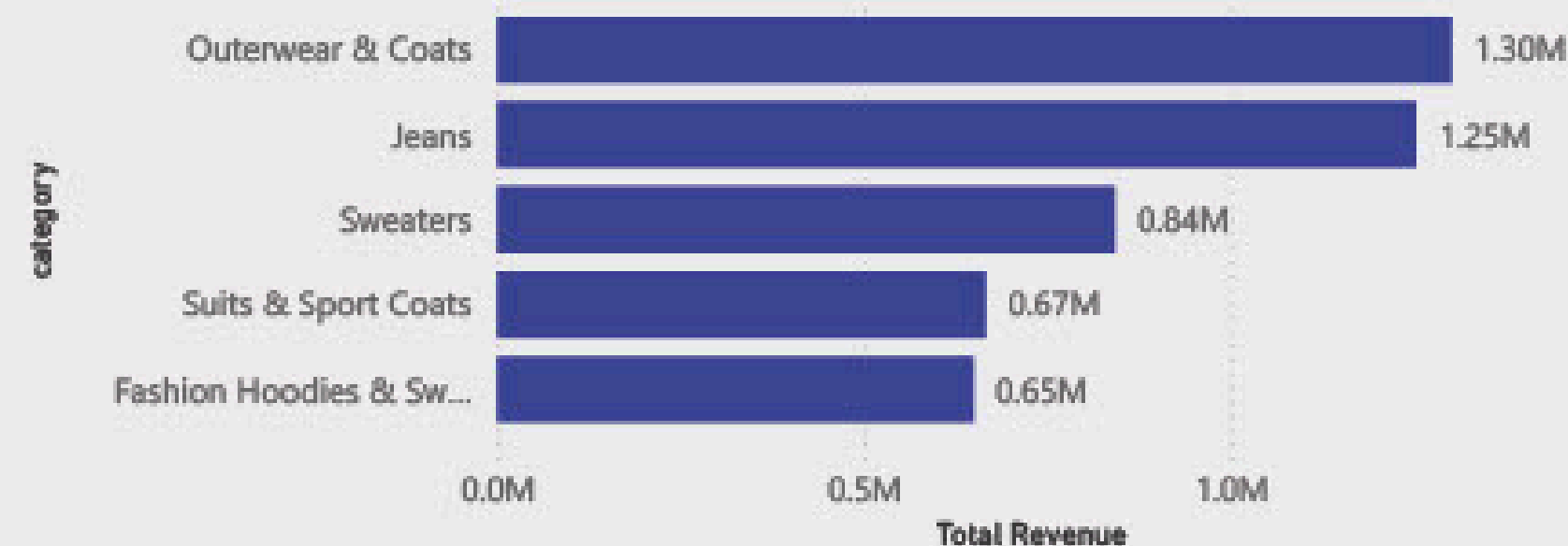
Monthly Revenue Trend



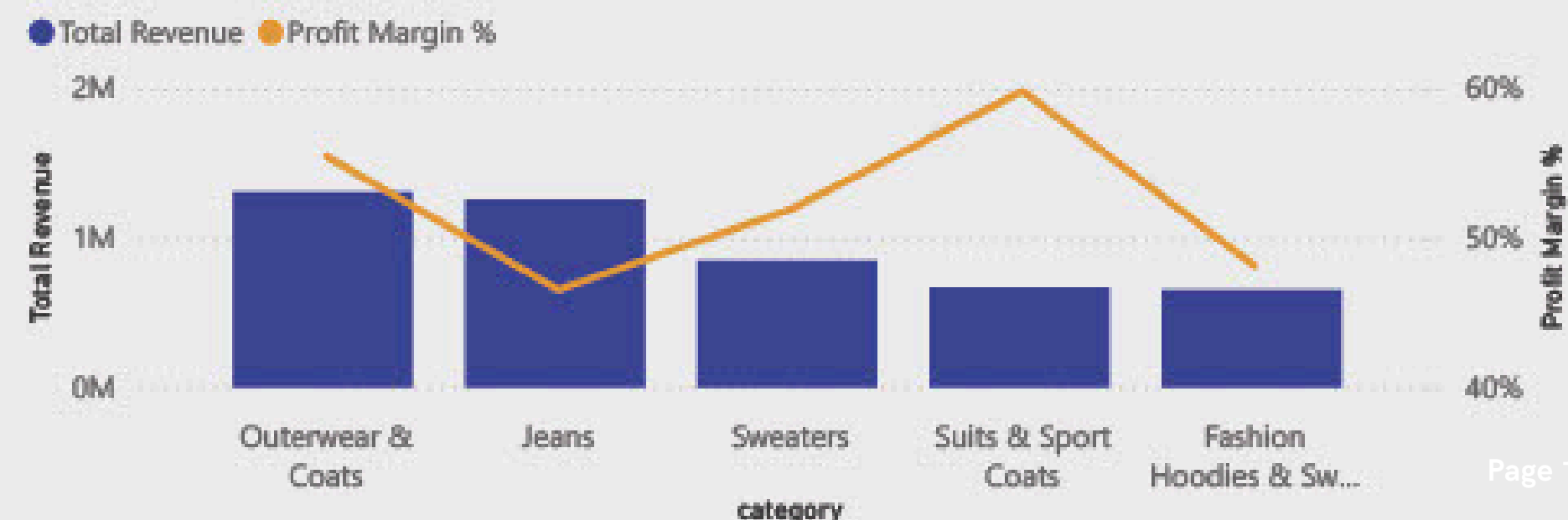
Monthly Orders & Active Customers



Top 5 Revenue by Category



Revenue & Profit Margin by Category



TheLook E-commerce Business Performance Analysis

Customer, Operational, Marketing Insight

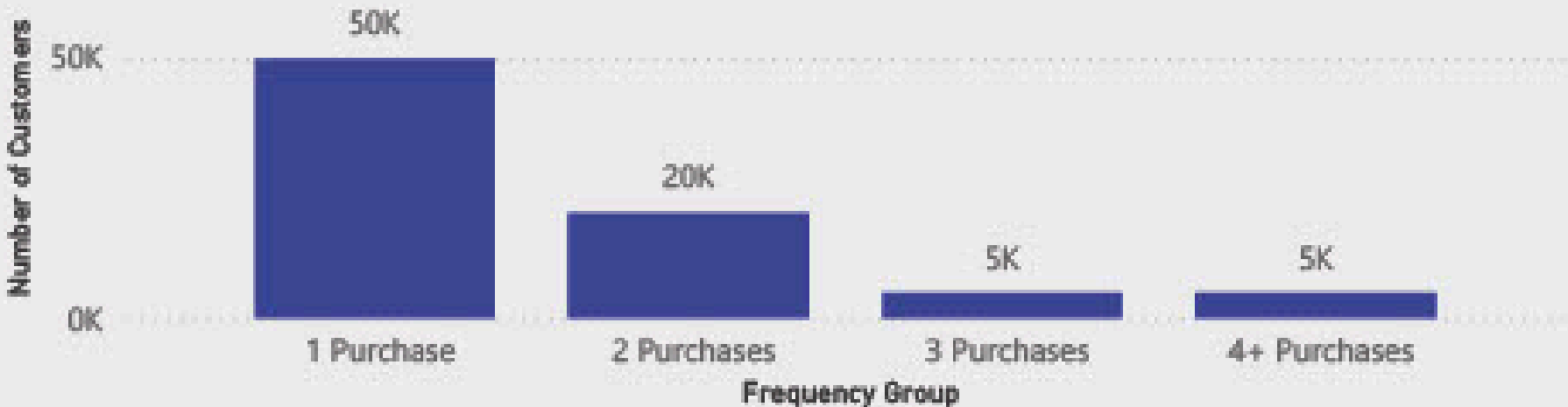
Year

All

Traffic Source

All

Customer Distribution by Transaction Frequency



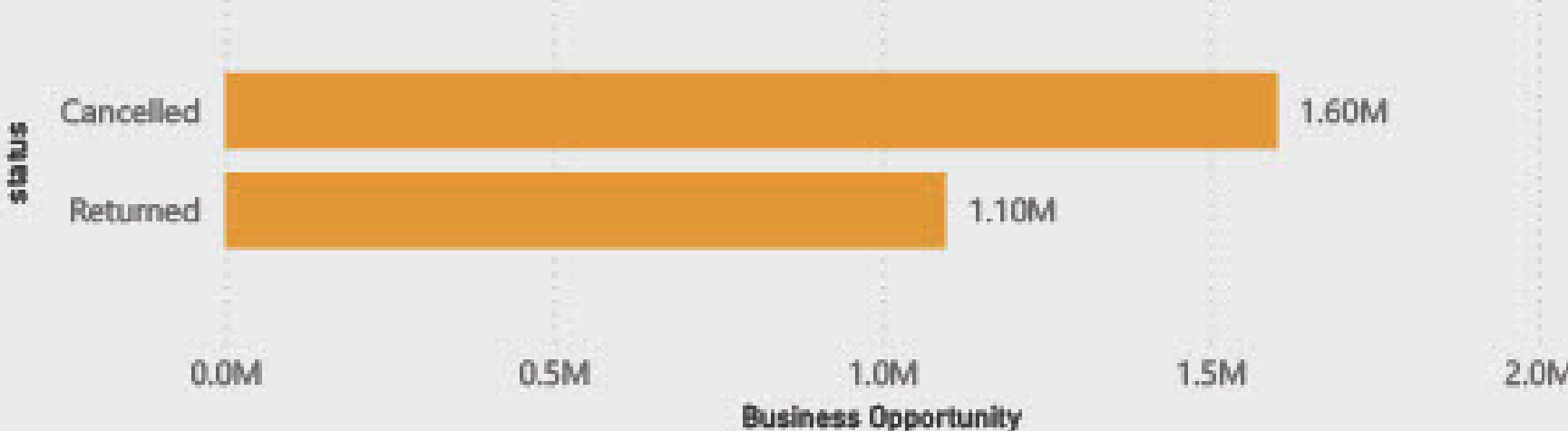
Average Revenue per Customer by Transaction Frequency



Order Status Distribution



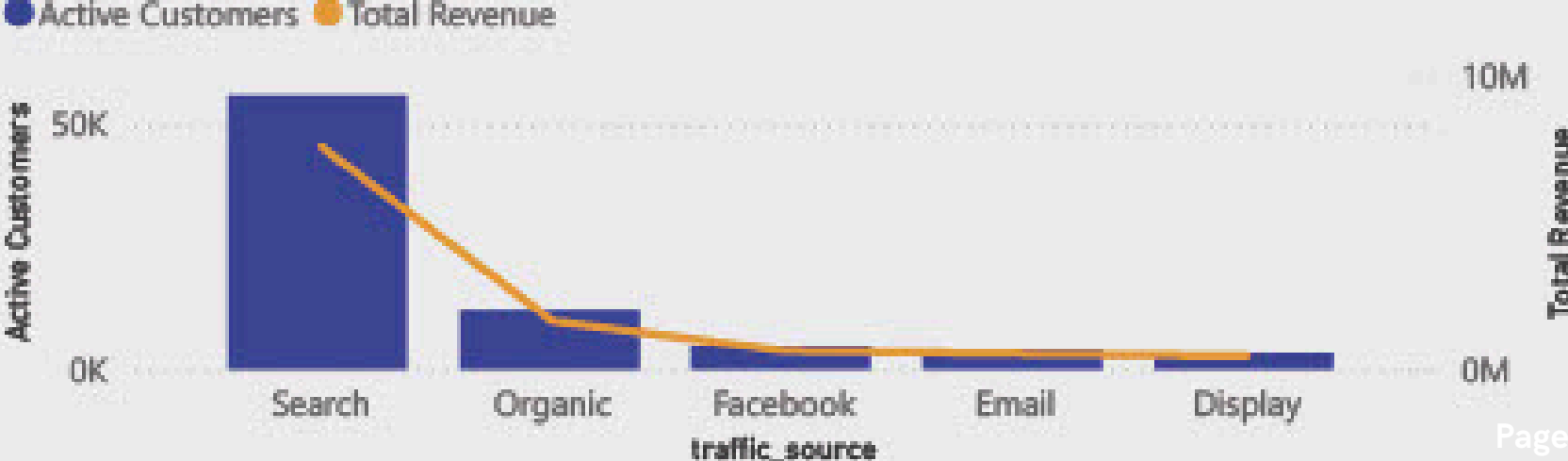
Business Opportunity Loss by Order Status



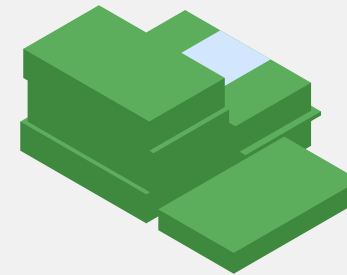
Revenue by Traffic Source



Active Customers & Revenue by Traffic Source



Recommendations



Increase Business Growth Through Transaction Expansion	Prioritize High-Performing Product Categories	Improve Customer Retention
Recommendation Increase transaction volume through customer acquisition, seasonal campaigns, cross-selling, and product bundling. Expected Business Impact • Increase transaction volume. • Drive sustainable revenue growth. • Strengthen long-term business growth.	Recommendation Prioritize high-performing product categories in inventory planning, merchandising, and promotional strategies while evaluating low-performing categories for improvement. Expected Business Impact • Increase profitability. • Optimize inventory management. • Maximize revenue contribution from top-performing categories.	Recommendation Implement loyalty programs, personalized promotions, email marketing, and post-purchase incentives to encourage repeat purchases. Expected Business Impact • Increase Customer Lifetime Value (CLV). • Improve repeat purchase rates. • Reduce dependence on new customer acquisition.

Recommendations



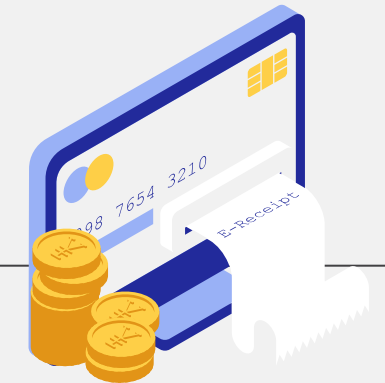
Reduce Order Cancellations

Recommendation

Identify the root causes of order cancellations and optimize payment, inventory, and checkout processes to improve operational efficiency.

Expected Business Impact

- Reduce order cancellation rates.
- Minimize potential revenue loss.
- Improve operational efficiency and customer satisfaction.



Optimize Search Marketing

Recommendation

Strengthen SEO and SEM strategies while continuously monitoring campaign performance to maximize Search as the primary acquisition channel.

Expected Business Impact

- Increase customer acquisition.
- Strengthen revenue contribution from Search.
- Improve marketing return on investment (ROI).

Project Summary

- Conducted an end-to-end business performance analysis using Python and Power BI to evaluate customer, product, marketing, and operational performance.
- Successfully identified the primary business growth driver, high-performing product categories, customer retention opportunities, operational bottlenecks, and the most effective marketing channel through five business-focused analyses.
- Delivered interactive dashboards and strategic recommendations that enable TheLook to improve business growth, customer retention, operational efficiency, and marketing effectiveness through data-driven decision-making.



Thank You

 [GitHub Repository.](#)

